

Acquisition Implementation

Learning Objectives

Awareness/Knowledge:

- Land survey principles
- Marine navigation and positioning
- Land and marine source operations
- Digital recording systems
- Quality assurance

Module 8

Acquisition Implementation

Why Important

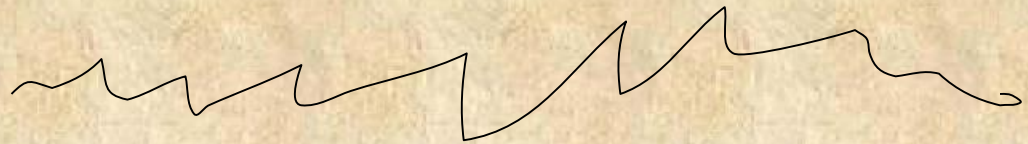
- Choices of sources and source parameters affect signal characteristics, quality, and processing control.
- Understanding the wavelet characteristics imposed by equipment aids interpretation of rock/fluid properties.
- Positioning, characteristics, and deployment of sources and receivers can differ from ideal designs.
- Knowledgeable quality control during acquisition ensures success.

What Is Good Seismic Positioning?

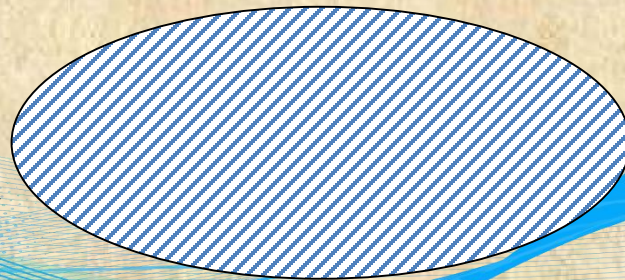
Good seismic positioning provides accurate horizontal and vertical positions for the entire underground feature.

One question that must be asked: **what will the positions be referred to ?**

Earth's Surface



Underground Feature
To Be Positioned



Target Reference.



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What is GPS and Why Do We Need It in Seismic Positioning?

The GPS (Global Positioning System) is a satellite-based positioning system which can be used night and day from virtually all corners of this planet.

The benefits of GPS in seismic work:

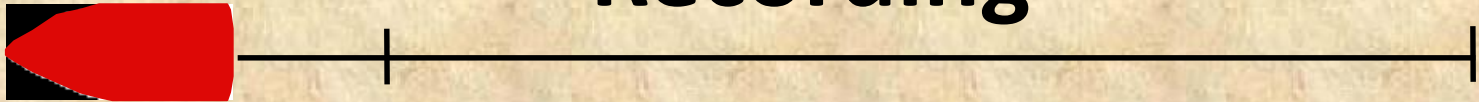
1. All-weather operation,
2. Extremely good accuracy over long distances,
3. 24 hour per day availability virtually everywhere,
4. Very good reliability,
5. Much faster than conventional surveying techniques in many situations. (e.g. real time source positions)

Sources and Receivers

Positioning Accuracy Requirements

	<u>Desired</u>	<u>Actual</u>
• Shots	.1-1m	3m
• Receivers	.1-1m	3m

Source and Receiver Positioning Assumptions 2-D Recording



- Streamer straight
- Source and streamer follow boat track
- Geometry often assigned based on shotpoint and receiver group intervals without use of navigation data

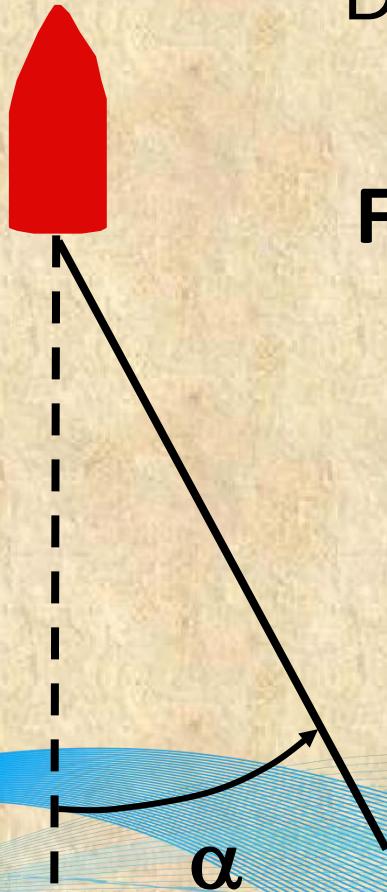
Cable Feathering Crossline

Cable Deflection

$$\text{Deflection} = \text{cable length} \times \sin(\alpha)$$

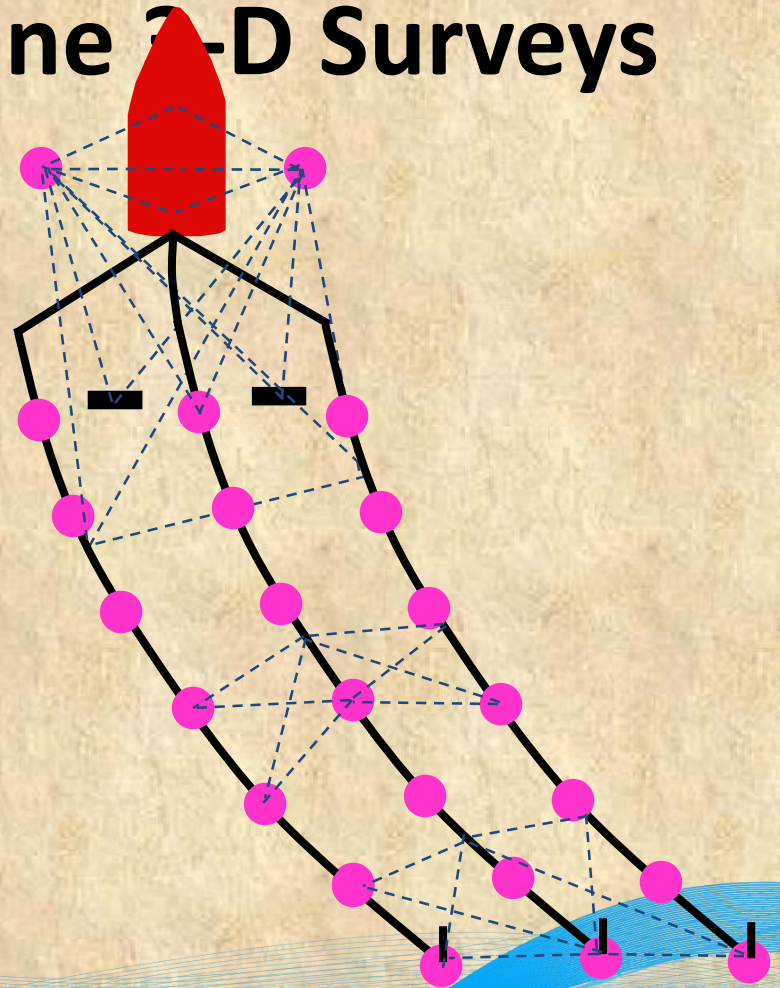
For a 6000 meter streamer:

1 degree	=	105 meters	
2 degrees	=	209 meters	
3 degrees	=	314 meters	
4 degrees	=	418 meters	= 1/4 mile
8 degrees	=	835 meters	= 1/2 mile
16 degrees	=	1654 meters	= 1 mile



Current Source and Receiver Positioning Marine 3-D Surveys

- Positioning measurements from a variety of sensors Including cable compasses, acoustic ranges, laser measurements, navigation buoy positions - are combined to form an Integrated positioning network
- Positioning accuracy indicators output along with coordinate estimates



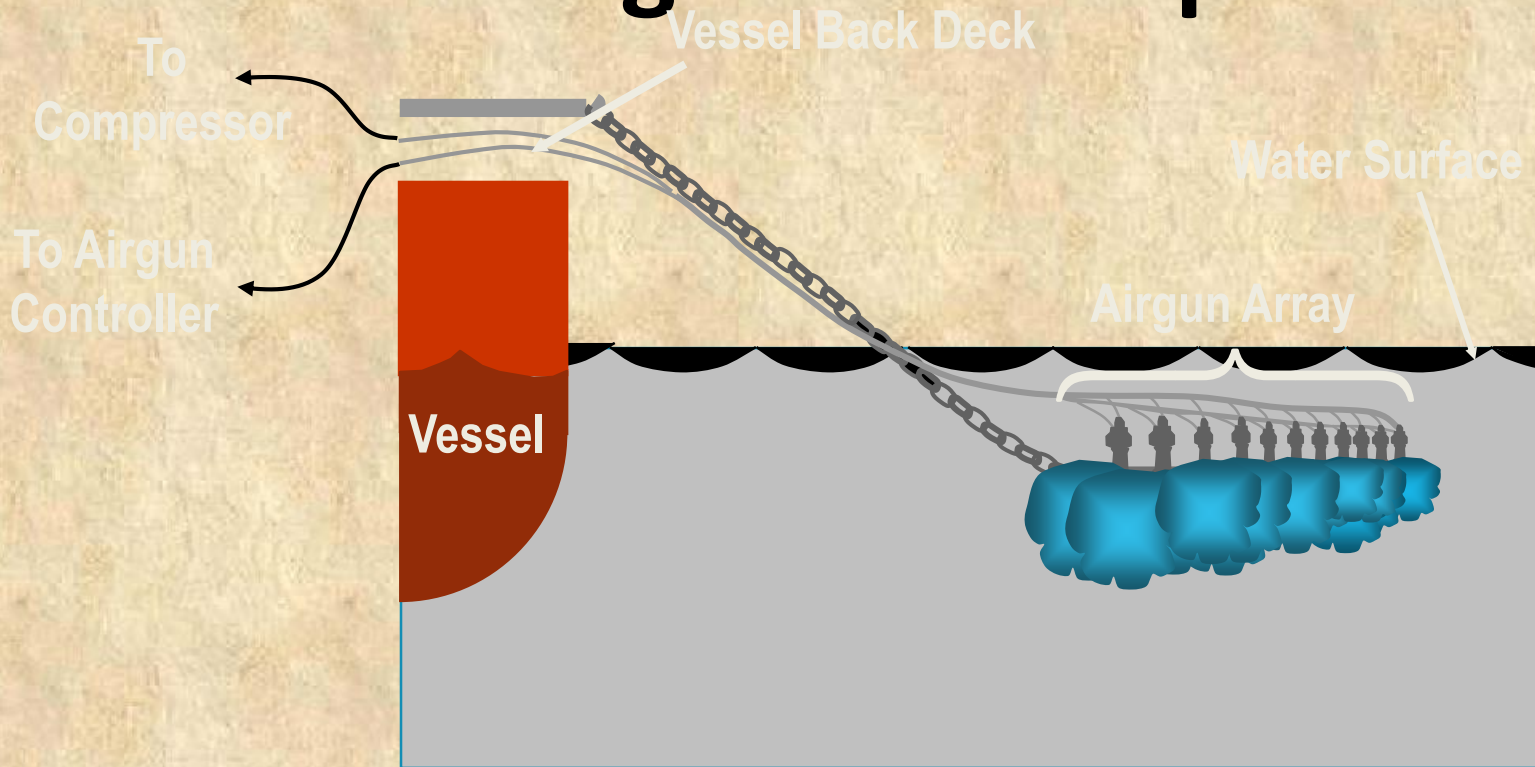
Seismic Energy Sources

- Marine
 - Air Gun, Water Guns, etc.
- Transition Zone
 - Dynamite in Shot Holes
 - Marine-Type Sources
- Land
 - Dynamite in Shot Hole, Vibroseis
- Source Selection Usually Based on:
 - Reliability
 - Ability to Generate Seismic Energy Required to Image Primary Objective
 - Cost Effectiveness
 - Accessibility
 - Repeatability

Marine Source Requirements

- Produce a powerful pulse
- Have capability of rapid firing (approximately 10 sec intervals)
- Be simple to operate, consistent and trouble-free
- Remain at constant depth and have minimum drag when towed
- Cause no injury to marine life
- Minimize bubble effect (repeated expansions and contractions of volume of gas)

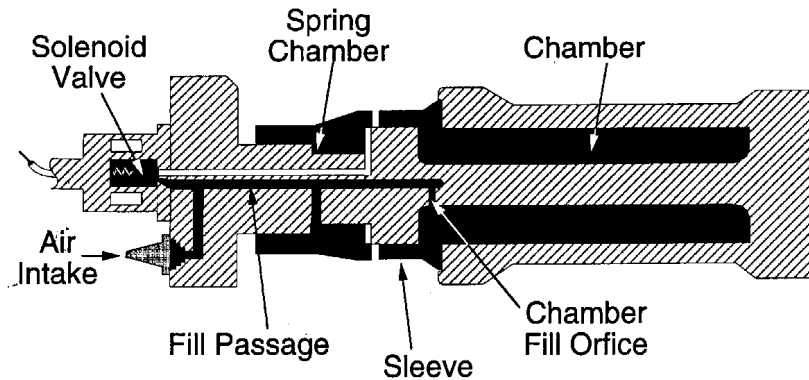
Airgun Technique



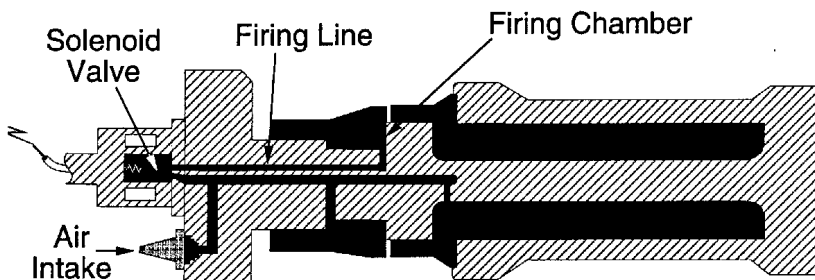
- Airguns of various sizes are combined to form arrays
- Compressed air is supplied by the compressor on the back deck
- Navigation system sends signal to Airgun Controller when source point is reached
- Airgun Controller fires each gun in array and sends signal to recording instruments to start recording
- Compressed air released by each gun forms bubbles which coalesce into a single bubble that provides desired waveform

Air Gun Operation

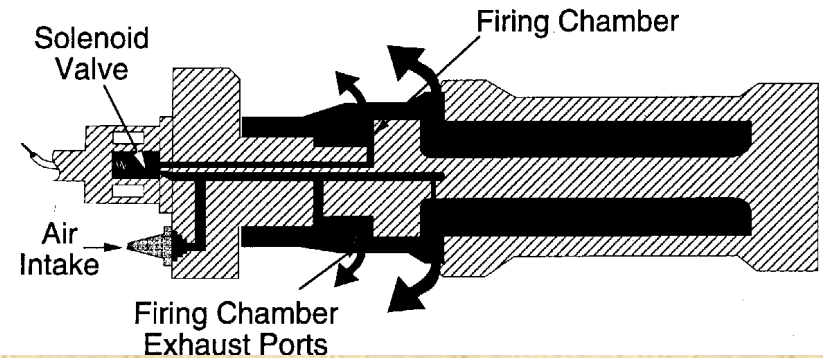
1. CHARGED



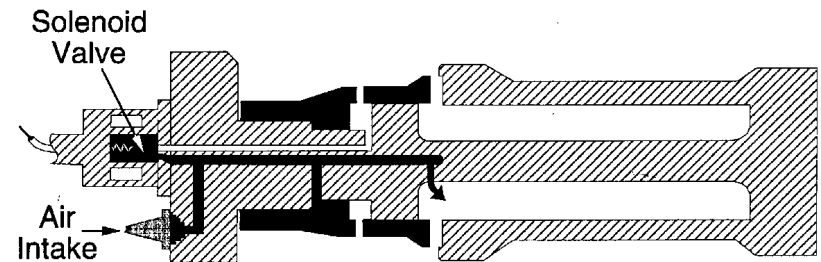
2. FIRED



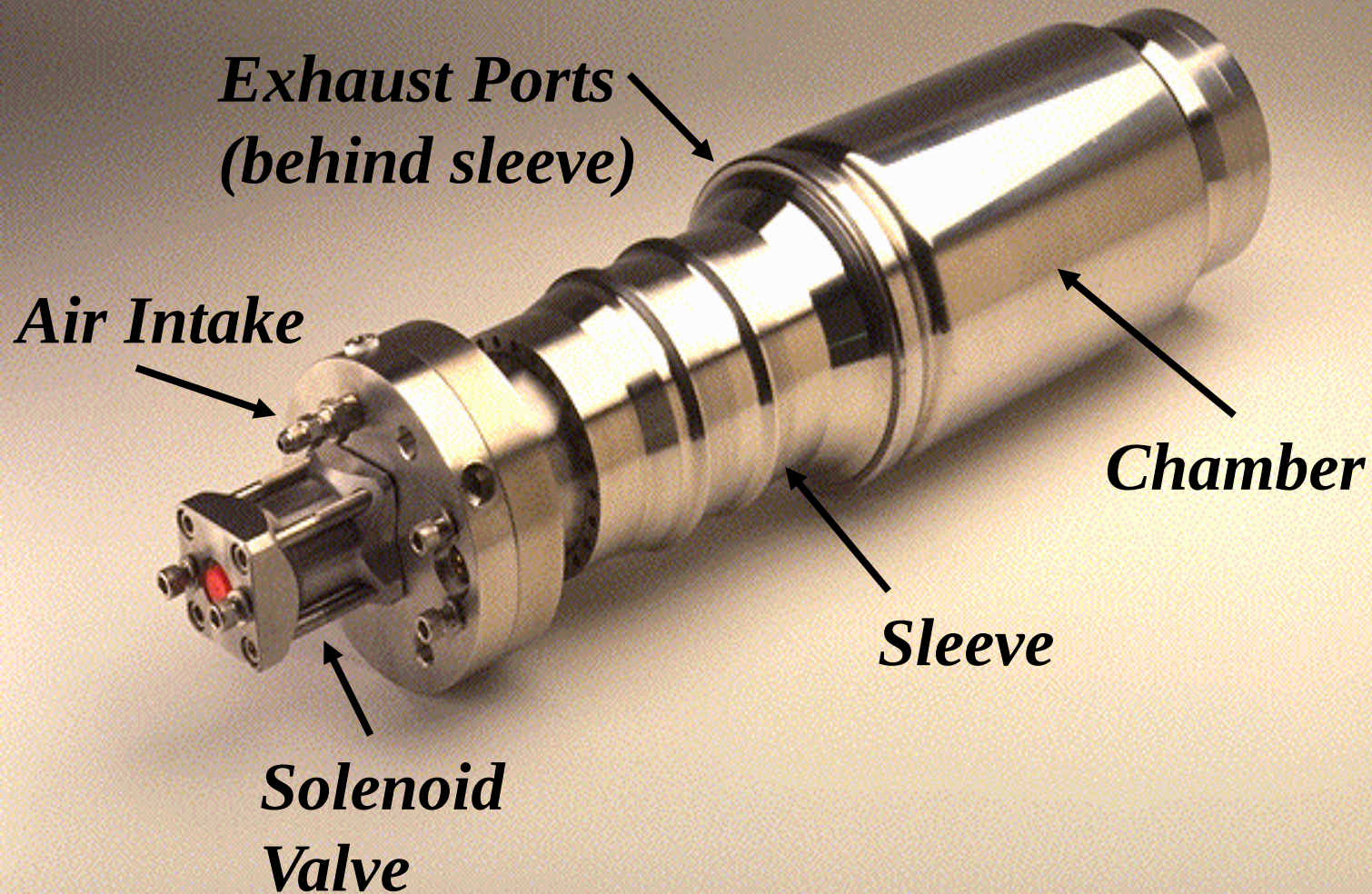
3. EXHAUSTING



4. EXHAUSTED

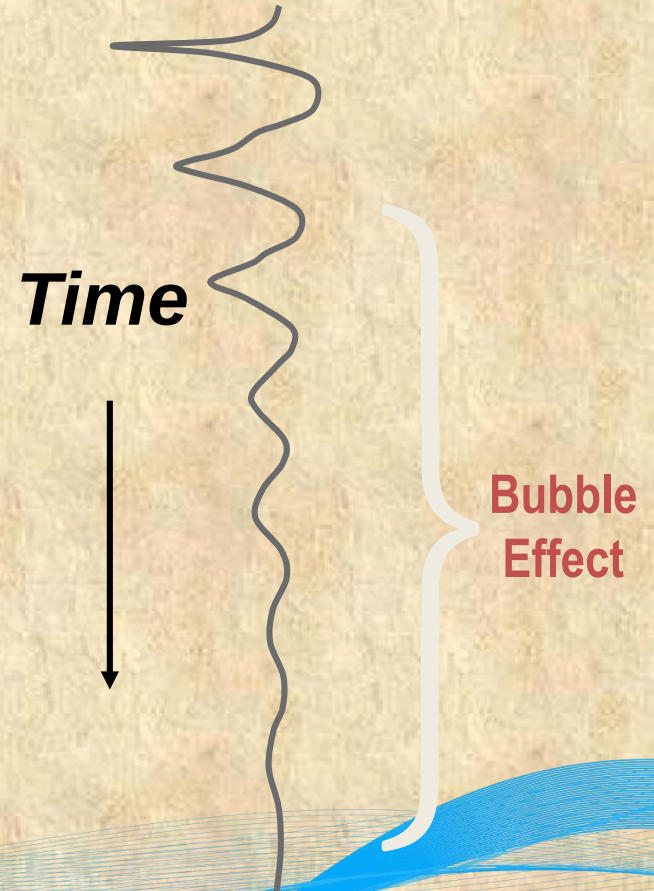
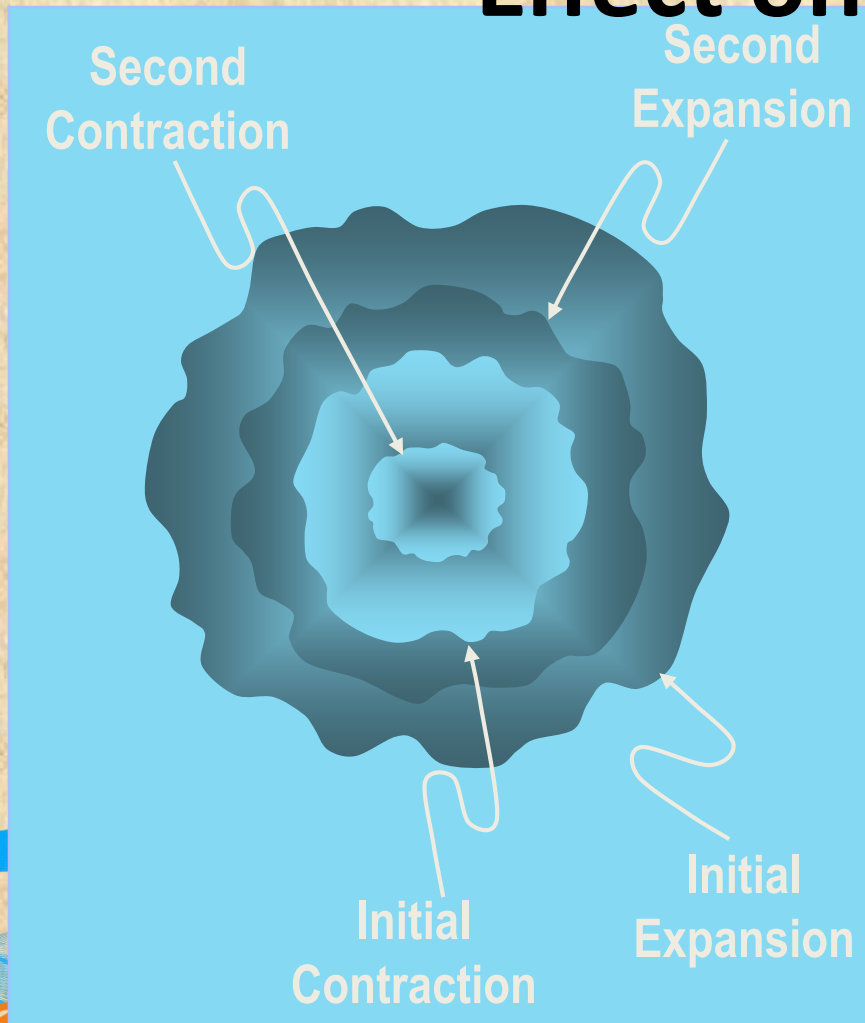


Sleeve Gun



Airgun Bubble Action and

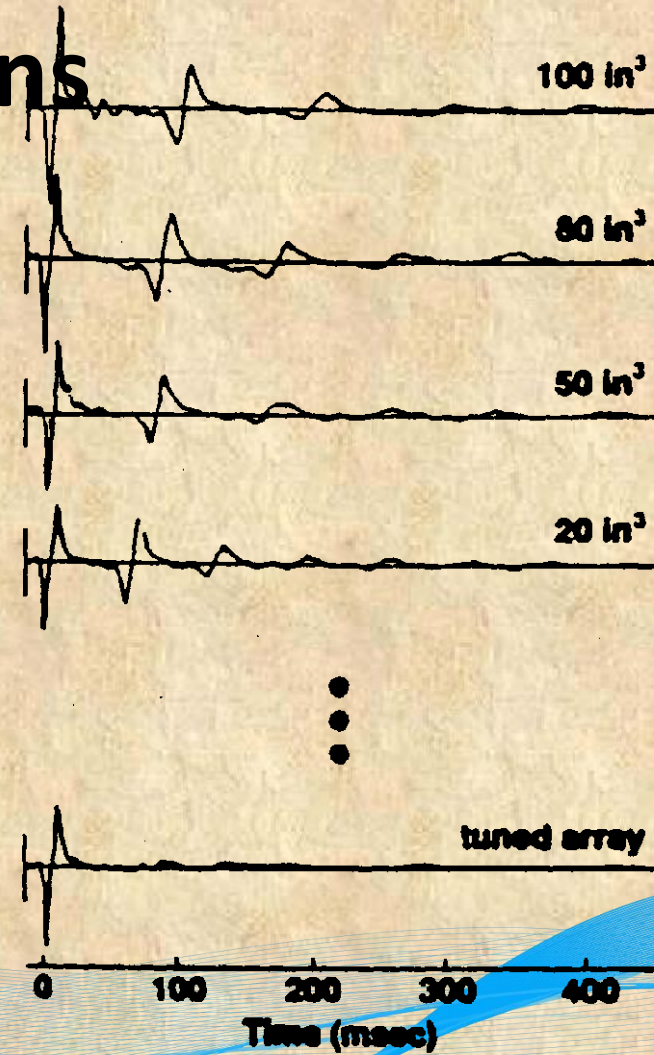
Bubble Effect on Waveform



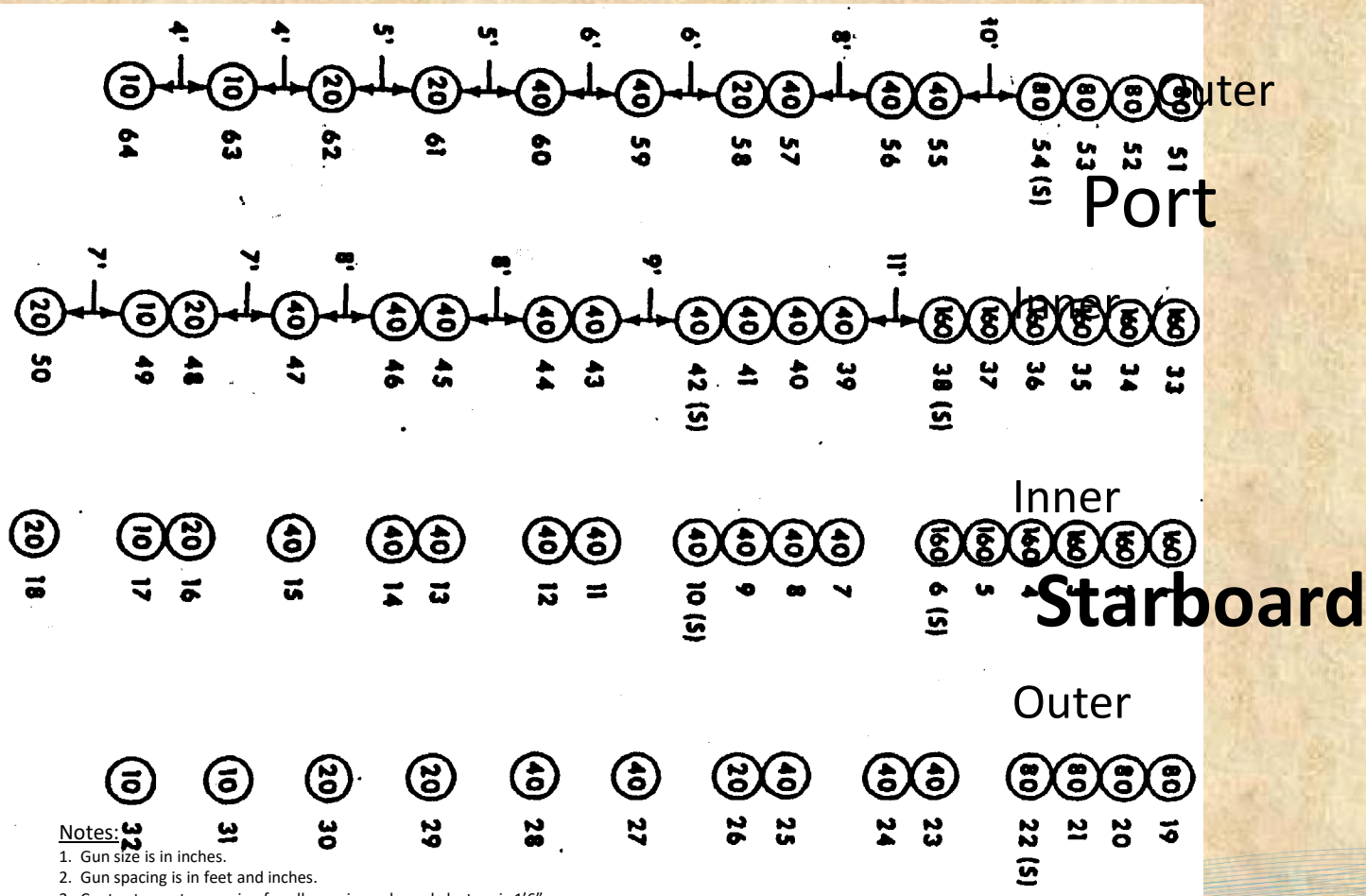
Signal Characteristics

Air Guns

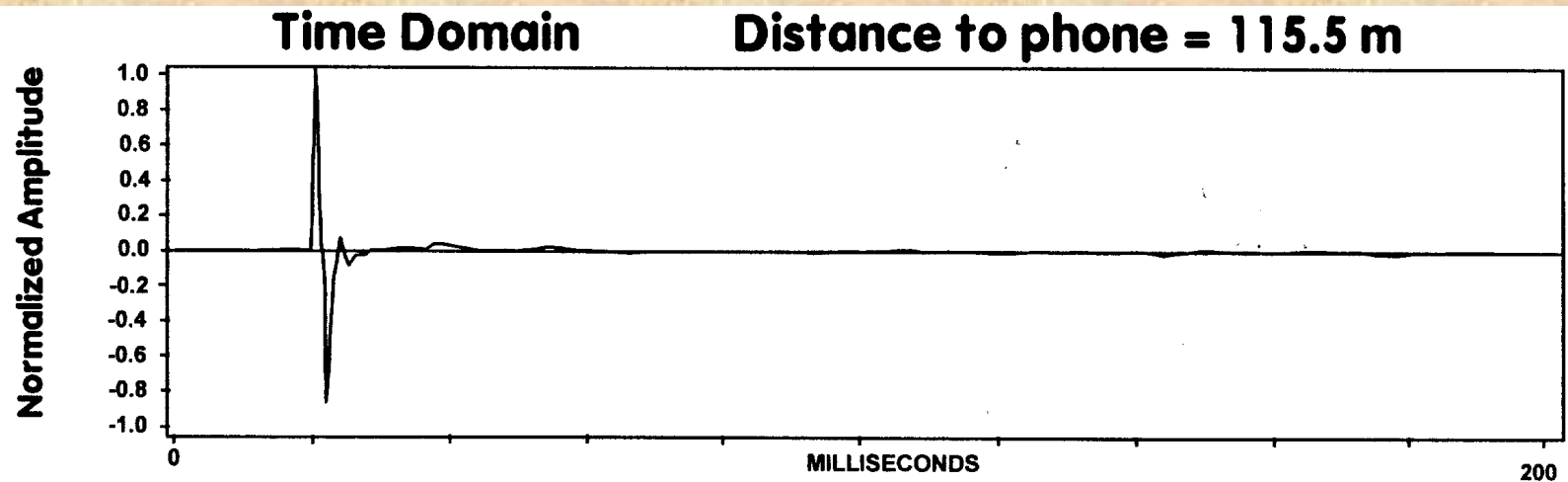
- To improve wavelet shape combine guns of various sizes into an array such that the initial expansions are in phase and the later ones (bubble effect) are out of phase (see bottom right and following slides).



3380 Cubic Inch Air Gun Array

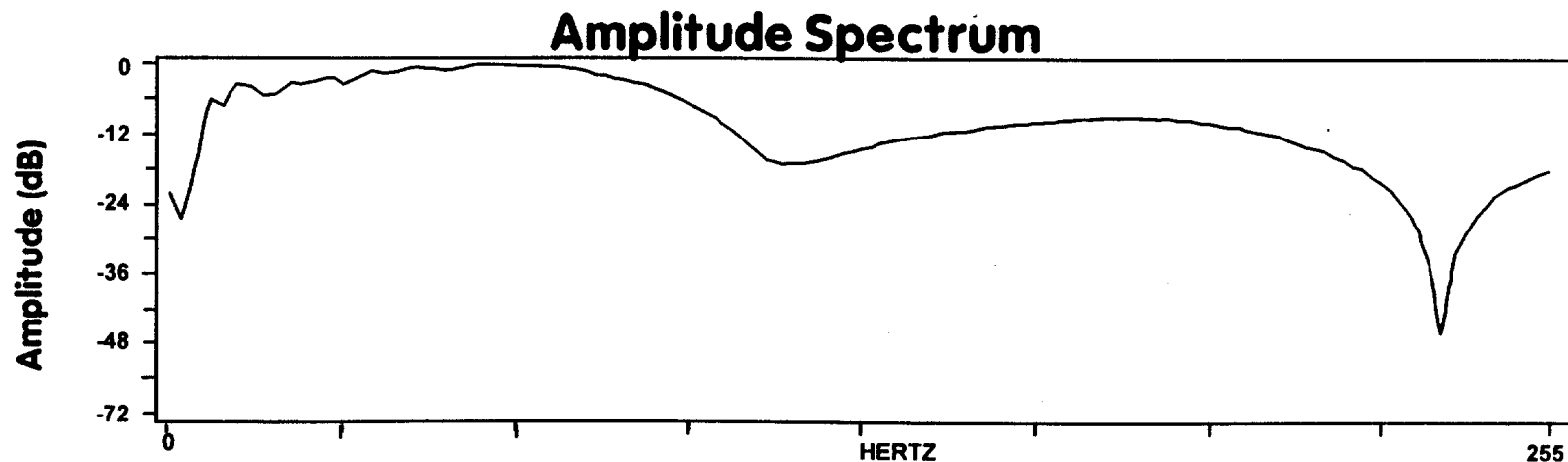


Signature of 3380 Cu-In Airgun



Normalized Amplitude: 1 = 1.49 volts

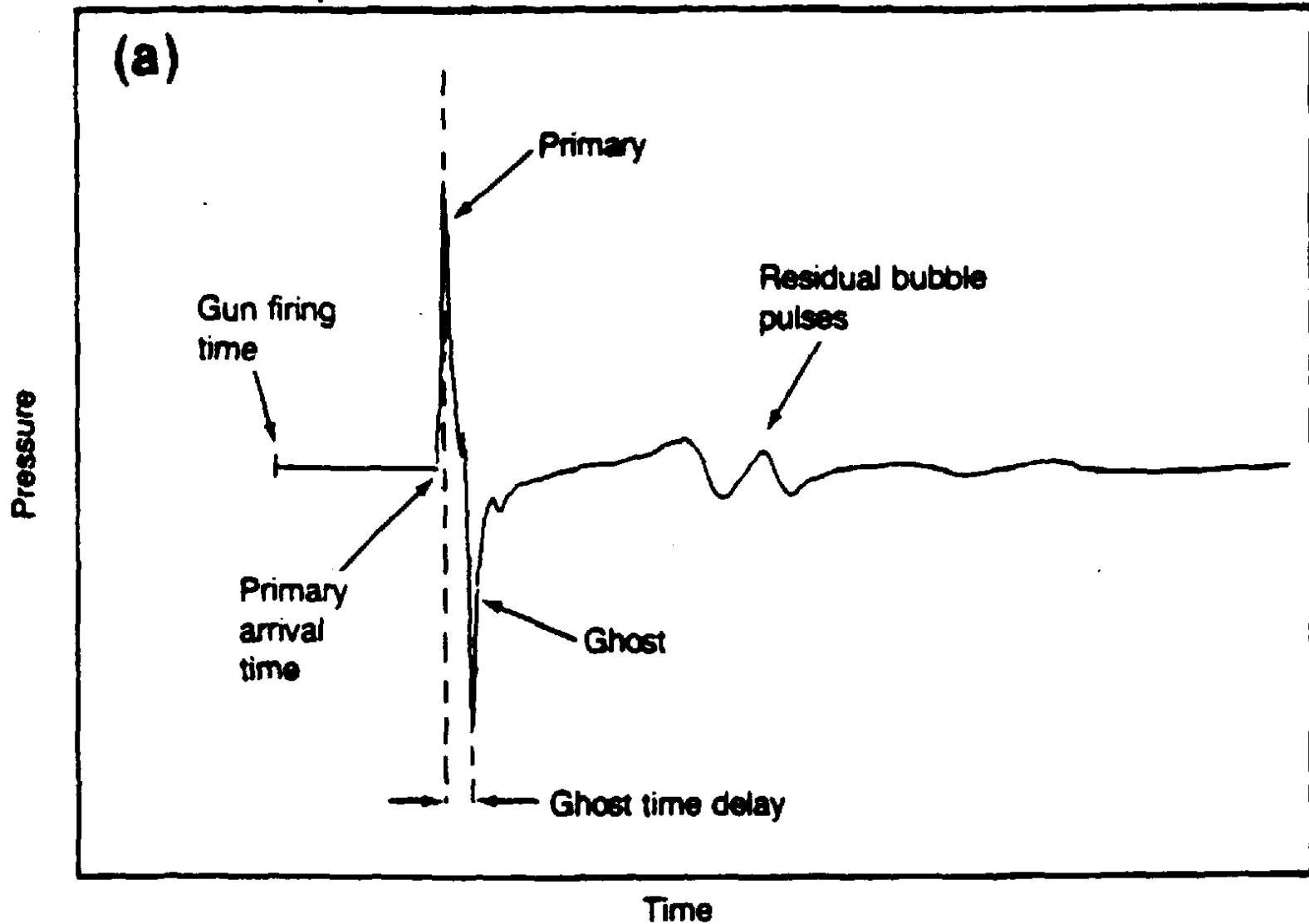
Sample Period = 2 ms



Foldover Frequency = 250 Hz



Time Domain Attributes



Land Energy Sources

- Normal choice is between vibrators and explosives. Choice depends upon:
 - Suitability to terrain, surface conditions, ease of drilling, availability, of personnel and equipment, environmental and governmental regulations, cost comparisons, quality of reflective energy.
- Explosives are not used in offshore data acquisition because of environmental

Land Source Parameters

Parameters for explosive sources:

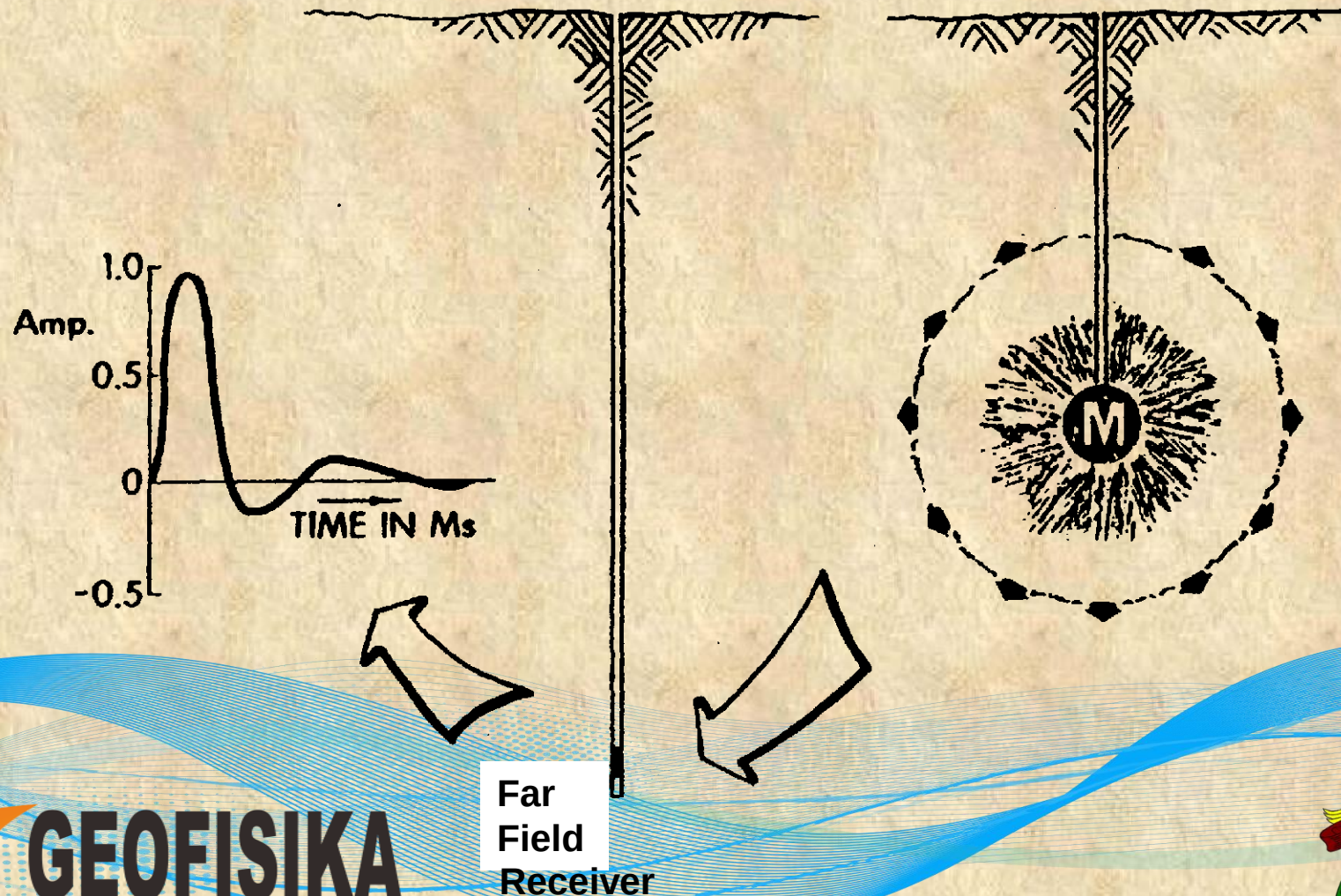
- Hole depth
- Number and pattern of holes
- Charge size

Parameters for vibrators:

- Number of units and size
- Source array and dimensions
- Number of sweeps
- Sweep type, start and end frequencies, sweep length and taper length

Explosive Sources

Measuring the far-field signature - Homogeneous Medium

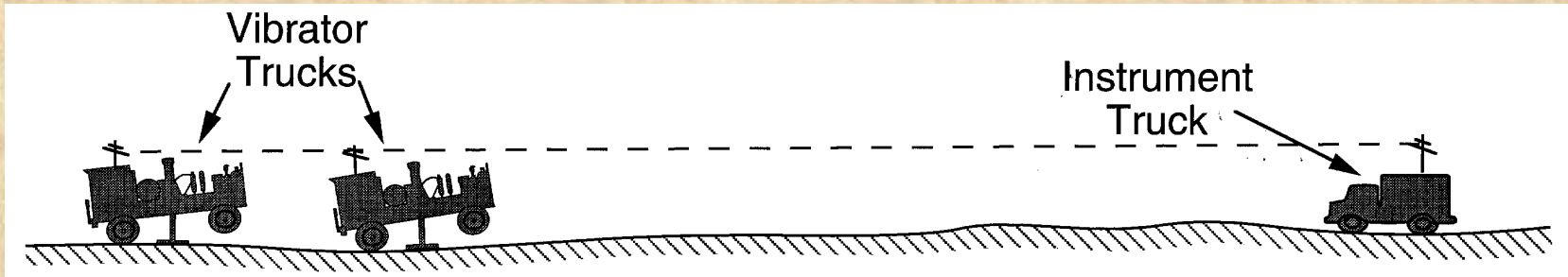


Explosive Sources

- Drilling deeper may place charges in a more favorable shot medium, producing higher amplitudes and wider bandwidth
- Field testing should indicate an optimum charge size and shot depth
- Drilling is one of the major costs in land seismic programs

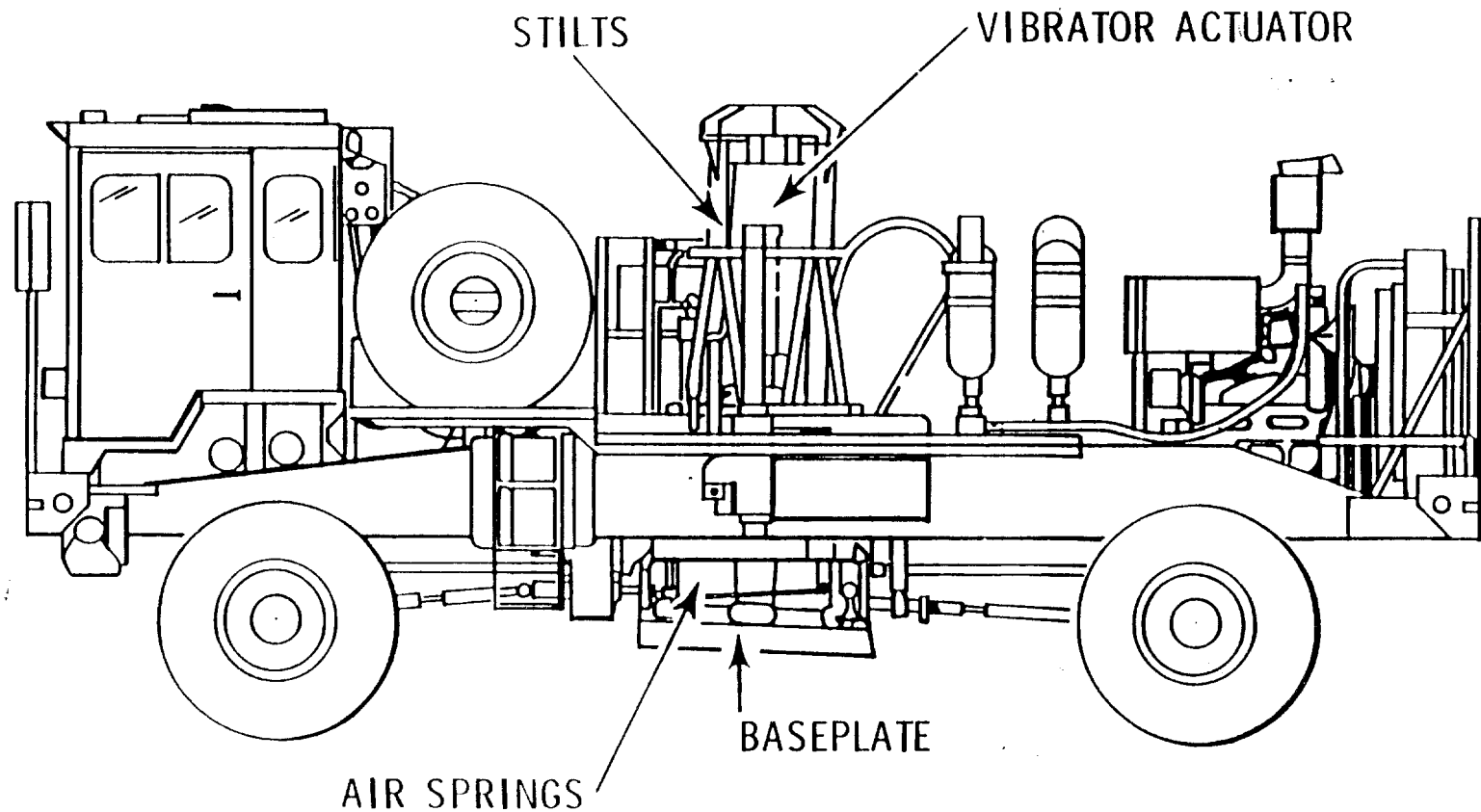


Vibrator Techniques



- Two or more vibrator trucks are positioned at source points within a source array (VP)
- The baseplates are lowered to the ground and the trucks are jacked-up to place additional weight on the baseplates, providing “reactive mass”
- An encoded swept-frequency signal (pilot sweep) is transmitted from the vibrator control unit in the instrument truck to similar units in each vibrator truck
- All vibrators input the prescribed signal into the ground and instruments begin recording simultaneously
- Recording continues for length of sweep plus “listen time”
- All records obtained at a single source point are vertically stacked (summed) into one record
- Records are cross-correlated with filtered pilot sweep to produce a record as long as the listen time

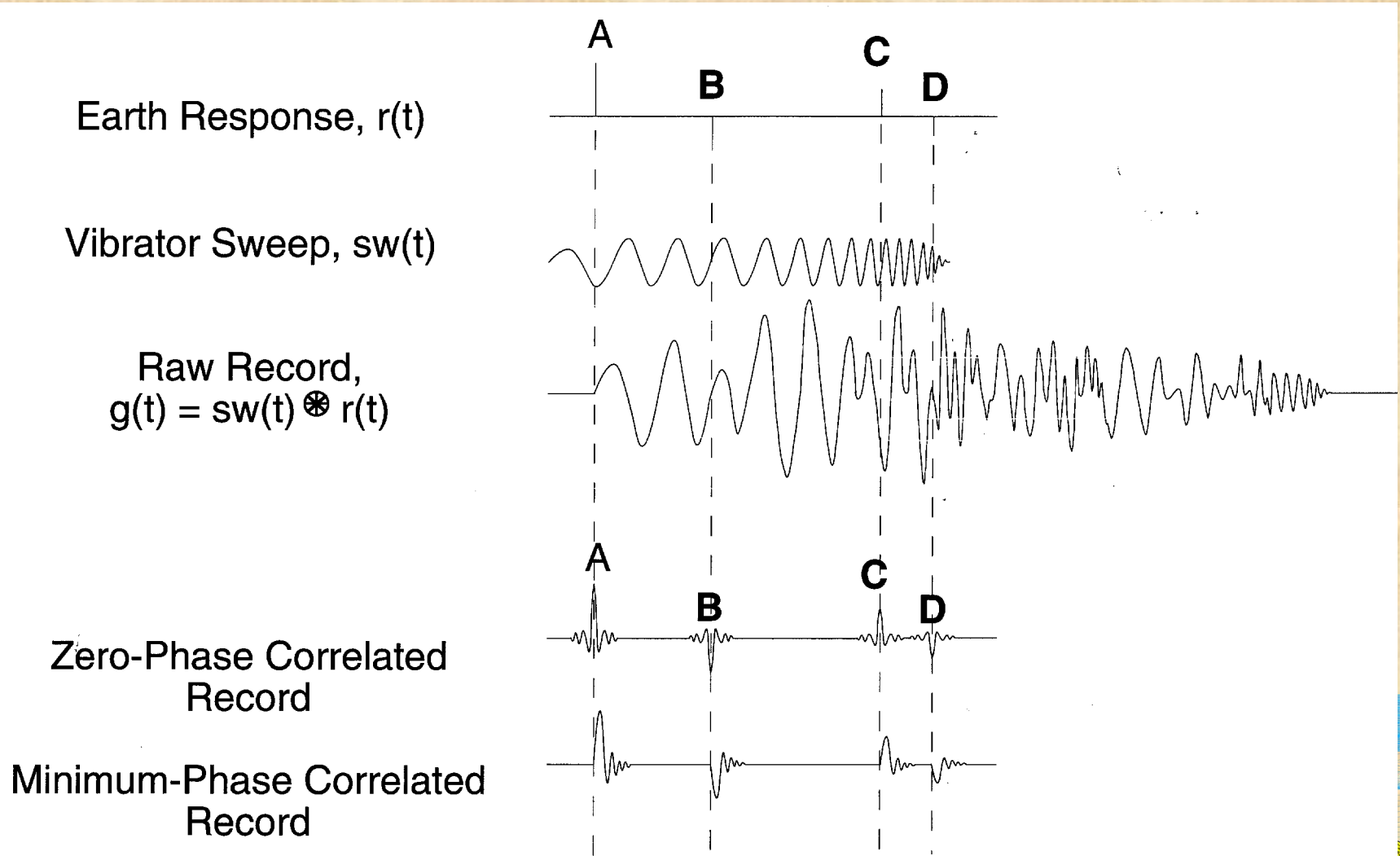
Truck-Mounted Vibrator Components



Side View of Truck-Mounted Vibrator



Raw and Correlated Vibroseis



Characteristics of Autocorrelation Wavelet

Resolution

- Resolution may be thought of as the width of the trough at the crossover point, i.e. - t_p = resolution
- *The above* assumes that all frequencies in a sweep are of the same amplitude and the sweep is linear.

Definition

- The ratio of the amplitude of the largest peak to the amplitude of the adjacent peak

Vibroiseis Signals

- Bandwidth of a vibrator is established by initial and final frequency values for the sweep
- Initial and ending tapers affect wavelet characteristics

Vibrator Summary

- Controlled frequency-content signal is produced. Maximum frequency range is from 5 Hz to 511 Hz. Maximum sweep length is 31 sec
- Wide variety of sweep types are available: linear, T^n , logarithmic, exponential, etc. Sweeps may be up (increasing frequency) or down (decreasing frequency). Cosine taper applied at each end of sweep to limit side-lobe amplitudes.
- Vibrator signal is low power but not low energy. Crosscorrelation of output record with pilot sweep collapses long duration signals into impulsive-appearing signals and increases signal power
- Two or more vibrator units used together. This increases cost but speeds operation. Multiple sweeps per shotpoint increase signal levels and signal-to-noise ratios.
- Vibrators can be used anywhere heavy vehicles can operate, including towns and cities, but cannot be used in marshy and mountainous areas or in jungles.
- Very hard surface may limit vibrators' effectiveness as this type of surface tends to distort signals.

Seismic Detectors



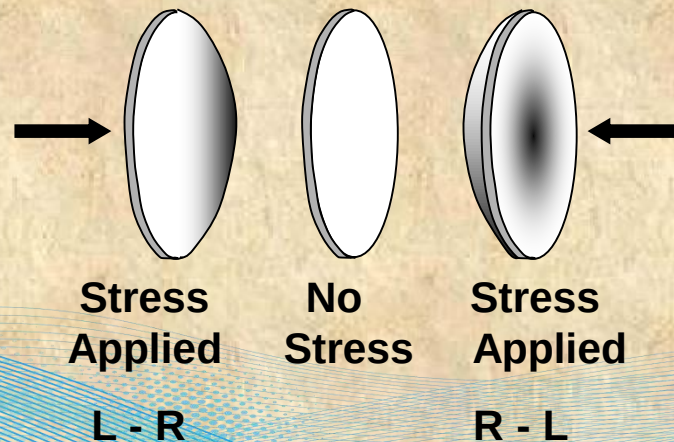
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Hydrophone

- In marine operations, hydrophones are under the water, and seismic energy reaches them as variations in water pressure
- Most hydrophones make use of the piezoelectric effect. Application of stress to the surface. In some materials, causes a flow of electrons in the direction of the stress.
- Since current is the flow of charge, a current is produced proportional to the applied stress
- Current across a load, produces a voltage



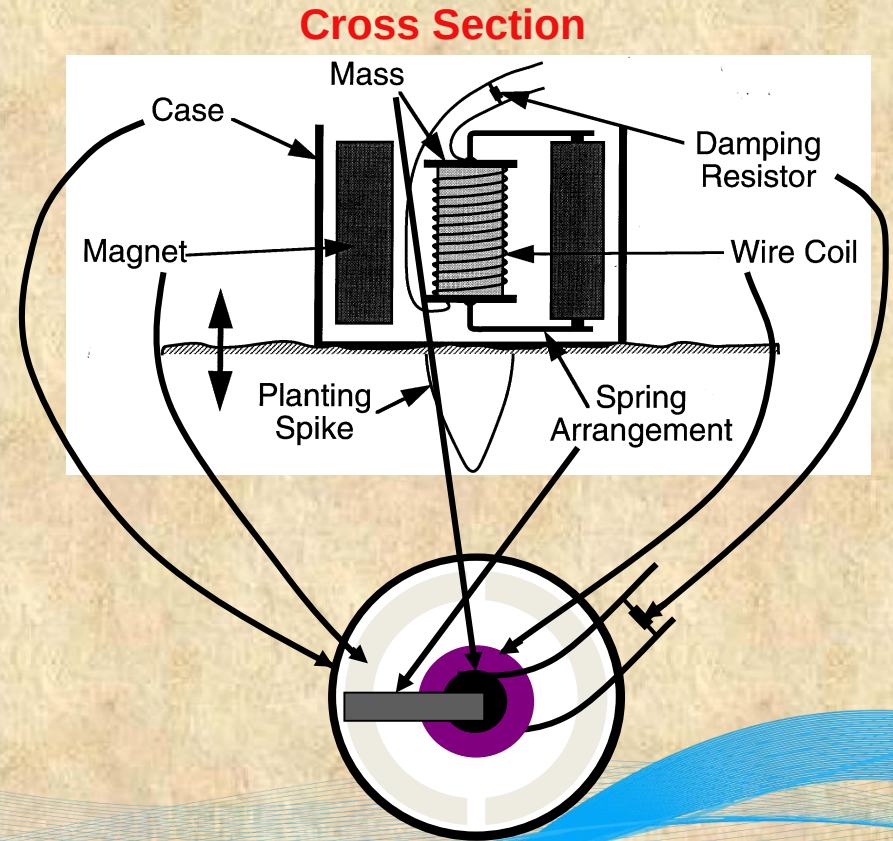
Piezoelectric Crystals

Geophone

- Shown at right is a simplified representation of an exploration seismometer, or geophone as it is more commonly called

- Principal components are:

- Case,
- Planting spike
- Permanent magnet,
- Mass,
- Wire coil
- Spring

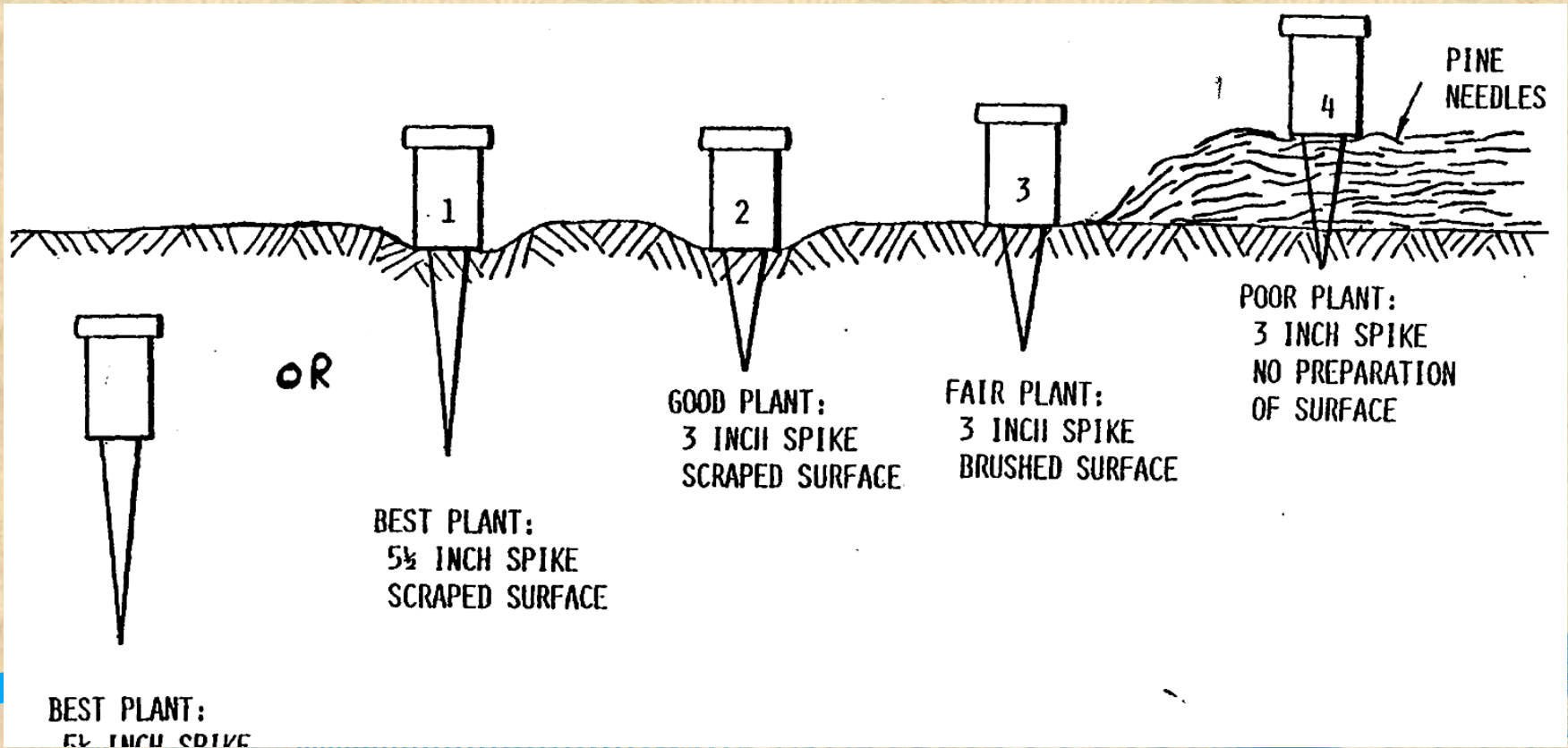


Top View
Geophone Components

Geophone

- The planting spike couples the case to the ground, so that we can assume that the case faithfully follows the ground motion. (Not always a good assumption).
- The magnet is attached to the case and moves with it
- The mass, with a wire coil wrapped around it, is suspended by a spring so that it remains fixed with regard to the earth as a whole
- Seismic waves arriving at the geophone, cause the case and magnet to move up and down

Geophone Plants



Seismic Recording Systems



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Distributed or Telemetry Recording Systems

- Modern seismic recording systems consist of two distinct parts
 - Ground (Land) or Inwater (Marine and OBC) systems
 - Truck or Onboard systems

Onboard or Truck-Mounted

- Major components is **System**
 - System Control Unit
 - Line/Streamer Interface Modules
 - Correlator/Stacker Module (optional land equipment used with Vibrators)
 - Operator Console Modules
 - Tape Transports
 - Printers and/or Cameras

Recording Room on Western Monarch



Quality Control



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Acquisition Quality Assurance for Interpreters

- Marine
- Land
- Safety and environmental audits are absolutely imperative

Sources and Receivers

Positioning Accuracy Requirements

	<u>Desired</u>	<u>Actual</u>
• Shots	.1-1m	3m
• Receivers	.1-1m	3m

QC Topics and Displays

Marine

1. Preplots of surface and image areas
2. Validation of basic parameters appropriate to meet survey objectives (sampling, binning, aperture, record length, etc.)
3. Far-field source measurement or estimation (variability with drop outs, etc.)
4. Coverage – all offsets, offset limiting, “flexing”
5. Areal RMS noise plots
6. Inline, xline, and cube stacks
7. Time slices
8. Streamer RMS noise

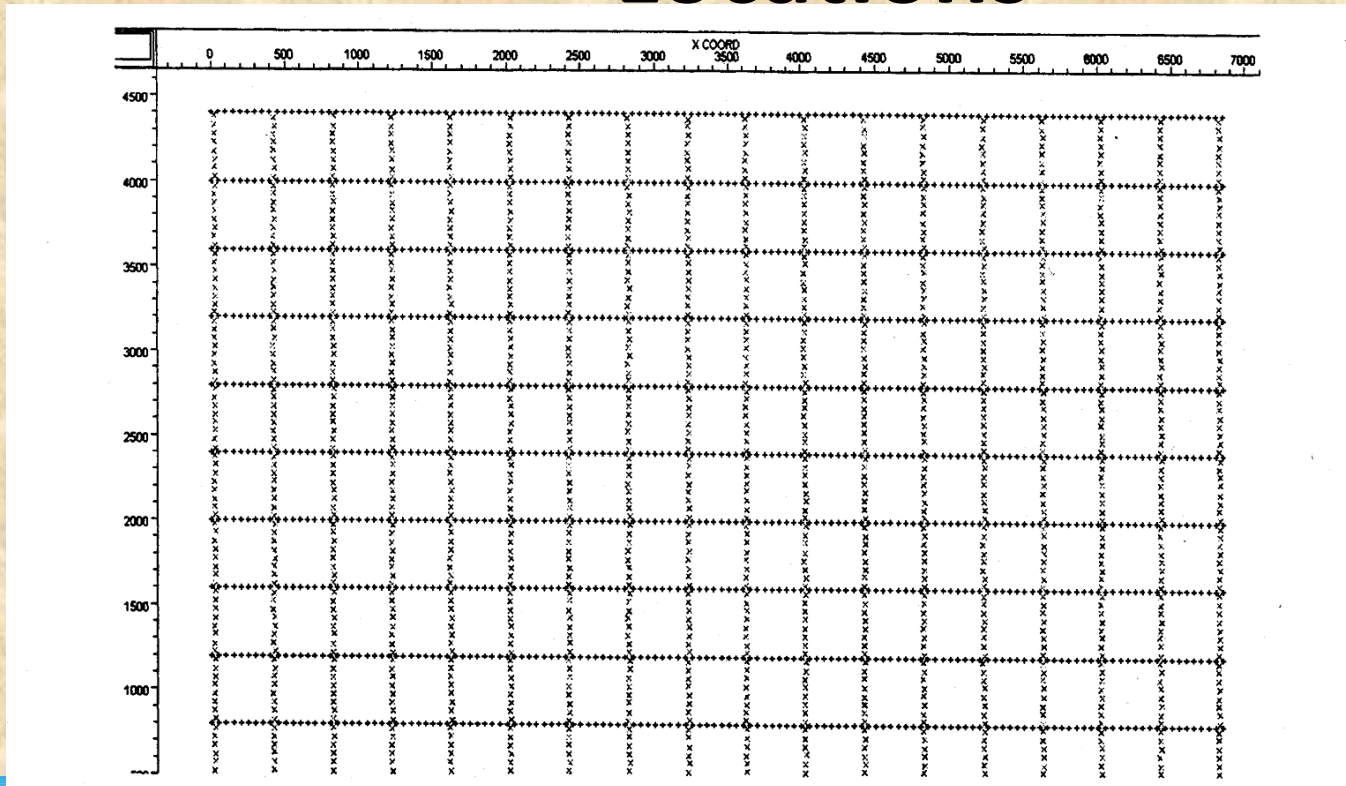
QC Topics and Displays

Land

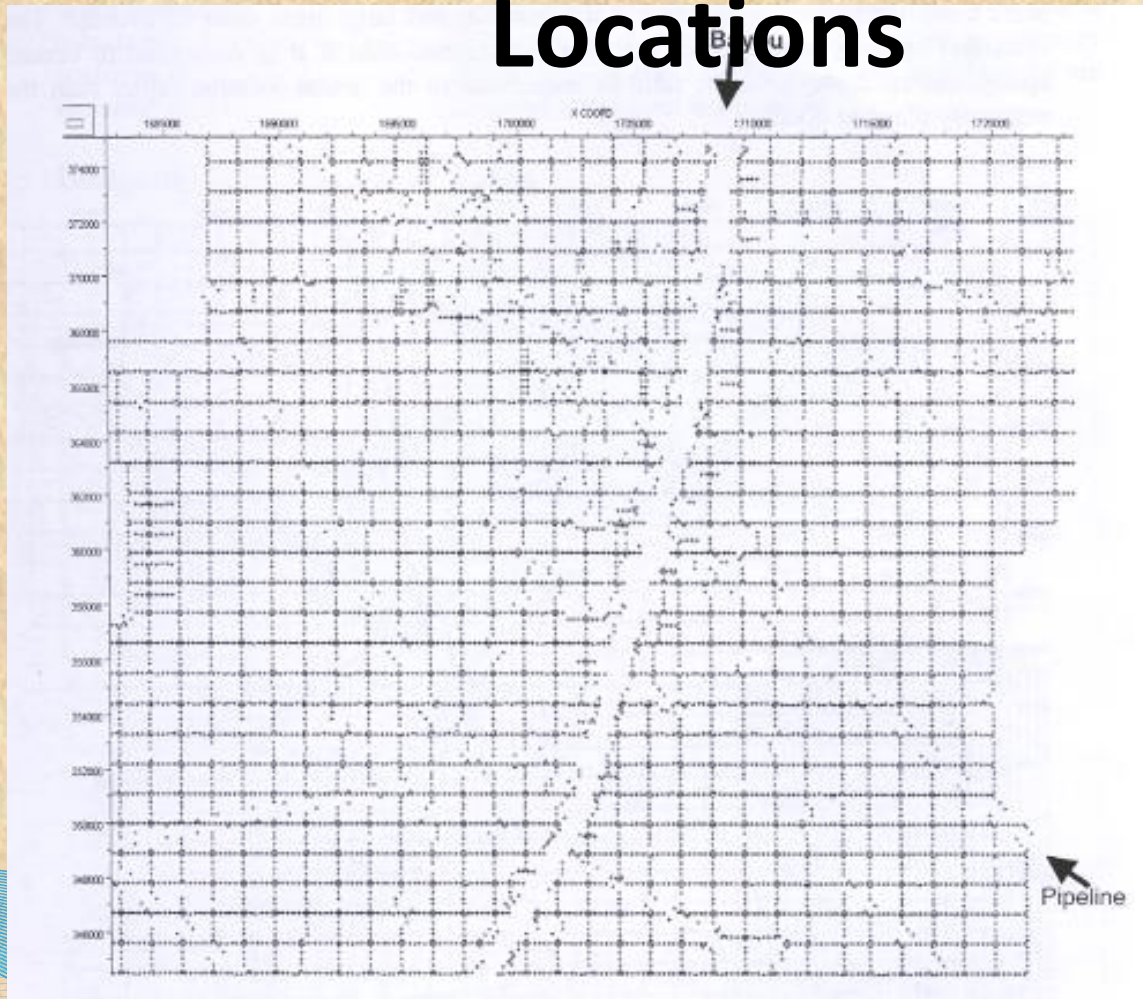
1. Preplots
2. Validation of parameters
3. Source estimation and variability
4. Actual coverage – offsets and fold
5. Shot records
6. Brute stacks selected lines
7. QC plots to identify geometry errors



Presurvey Source and Receiver Locations



Actual Source and Receiver Locations



Signature Estimation

- Direct measurement
 - Far Field
 - Near Field
- Modeling
 - Calibration with near field measurement
- Statistical estimation
- Wellbore analysis
 - VSP or synthetic
 - Correlate to seismic data

Quality Assurance at Crew

Start-Up Is Particularly Important

- For vibroseis
 - Sweeps and similarity tests
 - Polarity
 - Ground force measurement
 - Vibrator performance tests
 - Record signature
- For marine air gun
 - Pressure
 - Deployment of arrays and streamers
 - Recording polarities
 - Array performance



Summary

- What we learned about
 - Positioning requirements & accuracy
 - Marine cable feathering
 - Marine airgun arrays
 - Dynamite and vibroseis
 - Receiver characteristics
 - Quality control displays